

From Classroom to Enterprise: Architectural Patterns for LLM Integration

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Abstract—Large language models are rapidly diffusing into academic and enterprise knowledge work, yet most deployments lack explicit collaboration architectures, leading to inconsistent outcomes, unclear accountability, and unresolved risk exposure. Organizations respond with opposing approaches: academic institutions impose prohibitions or detection mechanisms to preserve the integrity of the author, while enterprises mandate usage to accelerate productivity. Both approaches fail because they regulate behavior without defining how humans and LLMs should collaborate within actual workflows.

This paper proposes centaur and cyborg frameworks as distinct human-LLM interaction patterns that differ in authority allocation, control boundaries, and failure modes. Centaur architectures maintain explicit decision boundaries with human authority over validation and final decisions, while cyborg architectures enable continuous collaboration with distributed oversight and supervisory control. We ground these frameworks in parallel longitudinal observations from academic (graduate data visualization course, seven semesters 2022-2025) and enterprise (software engineering and business operations) contexts, documenting remarkably similar evolution across three phases: novelty-driven atomic usage, stark shift to delegated patterns, and eventual differentiation into stable collaboration architectures. Both environments show atomic usage as the dominant pattern, with substantial subsets exhibiting centaur-aligned and degraded-centaur patterns, and small minorities demonstrating emergent cyborg usage.

The convergence of patterns across contexts with opposing governance mechanisms suggests that these architectures represent recurring patterns observed in contrasting contexts rather than context-specific adaptations. We translate these observations into practical deployment guidance that emphasizes maturity-gated adoption, selection of risk-tiered architecture, and lightweight embedded controls over enforcement-based policies. Contribution is an architectural framework for designing, evaluating, and governing LLM-enabled workflows grounded in observed collaboration behaviors.

Index Terms—Human-AI collaboration, large language models, collaboration architectures, governance frameworks, enterprise deployment, knowledge work

I. INTRODUCTION

Organizations adopt large language models expecting productivity gains but encounter inconsistent outcomes, unclear accountability, and unresolved risk exposure. Academic institutions respond with prohibition or detection, while enterprises mandate “use AI in your daily work” without specifying how collaboration should occur. Both fail because they regulate behavior without defining the collaboration architecture.

From a systems perspective, LLMs are active components whose outputs influence decisions and interpretations. Effective deployment requires explicit design choices about control boundaries, responsibility allocation, and error containment. Without architectural specification, emergent behaviors develop that may misalign with organizational risk tolerance.

This paper makes three contributions of decreasing scope. The primary contribution is a formal architectural framework distinguishing centaur (gated validation control) from cyborg (supervisory control) patterns for human-LLM collaboration, grounded in systems engineering and HSI theory. **Centaur architectures** maintain an explicit separation between model computation and human judgment through enforced validation gates. **Cyborg architectures** integrate LLMs continuously through supervisory control, with humans monitoring and intervening by exception. In support of this, we present illustrative case observations from academic and enterprise contexts. Finally, we translate the framework into deployment guidance through maturity-gated adoption, risk-tier alignment, and systematic FMEA that quantify failure modes.

II. BACKGROUND

Research on human-AI collaboration has emphasized human-in-the-loop (HITL) designs where automated systems generate output for human review. Such models work for bounded tasks, but degrade as automation becomes pervasive. The design challenge is not whether humans are “in the loop” but how authority and error detection are distributed [1], [2]. Prior work shows approval-based HITL designs degrade when oversight shifts from active judgment to nominal review. An integrated analysis of complacency and automation bias demonstrates that these are overlapping attentional phenomena that affect novice and expert operators, resist training interventions, and worsen under multitask load [3]. The resulting failure modes—misuse, disuse, and abuse of automation—arise from system design choices rather than operator incompetence [4].

A systematic review of 35 peer-reviewed studies spanning cognitive psychology, human factors, HCI, and neuroscience (2015–2025) finds that trust and task complexity are primary drivers of overreliance, and that neither accountability mandates nor explanatory interfaces reliably mitigate automation

bias across contexts [5]. Instead, **trust and task complexity** emerge as primary drivers of over-reliance, where the high **cognitive load of verification** often leads to **degraded oversight**—a failure mode where review becomes perfunctory rather than functional. These findings demonstrate that behavioral interventions alone are inadequate, necessitating **architectural controls** that technically enforce human authority.

Academic responses to generative AI have emphasized prohibition or detection to preserve the integrity of the author. However, detectors fail to generalize and are easily bypassed [6], disproportionately flagging non-native English speakers [7]. Systems safety research offers complementary insights: safety and reliability are emergent properties of system architecture, not achievable through component-level intervention alone [8]. The foundational observation that automating routine tasks paradoxically makes remaining human tasks harder—by degrading skills, reducing engagement, and creating monitoring burdens—has persisted across four decades of automation research [9] and directly motivates the architectural approach adopted here.

III. COLLABORATION ARCHITECTURES

Two distinct human-LLM collaboration architectures emerge from observed patterns across knowledge-work systems, each characterized by fundamentally different approaches to authority allocation, control boundaries, and failure behavior. We adopt Kasparov’s evocative terminology of ‘centaur’ and ‘cyborg’ architectures [10] while grounding them formally within systems engineering control frameworks.

From a control systems perspective, these represent fundamentally distinct approaches to human-automation interaction. Centaur architectures implement gated validation control, characterized by discrete decision boundaries where human authority is explicitly retained through structured review gates. Cyborg architectures implement supervisory control, characterized by continuous human oversight with dynamic authority allocation and exception-based intervention.

We use centaur and cyborg as accessible shorthand; the technical distinction is between gated validation control and supervisory control structures. This dual nomenclature enables both effective communication through evocative terms and systematic engineering evaluation through formal control frameworks.

A. Centaur Architecture

A centaur architecture features an explicit separation between model computation and human judgment, implementing gated validation control where decision authority concentrates at explicit checkpoints. The LLM generates candidate artifacts while humans retain authority over problem framing, validation, synthesis, and final action. Key characteristics: (1) human-owned problem definition with objectives and constraints set before model invocation; (2) model-generated proposals treated as provisional inputs; (3) explicit validation where humans evaluate outputs and integrate domain knowledge; (4) auditable decision points where transitions

from output to action are observable. When implemented correctly, centaur architectures balance productivity gains with risk containment, suiting regulatory, safety, or accountability environments.

The common failure mode is **degraded oversight**, where human review collapses into superficial approval under time pressure or repeated exposure, transferring authority to the model despite nominal control. This failure is architectural rather than individual: even experienced commercial pilots exhibit automation bias under operational conditions, defaulting to heuristic acceptance of automated recommendations regardless of expertise [11]. The failure arises when workflows lack enforced validation or when incentives reward speed over judgment.

B. Cyborg Architecture

A cyborg architecture integrates LLM capabilities continuously across workflows, implementing supervisory control—the paradigm in which humans set goals, monitor execution, and intervene by exception while automation handles routine execution [12]—where human authority operates through continuous monitoring and exception-based correction rather than discrete approval gates. The system may act by default while humans supervise, steer, and intervene by exception, with authority sharing dynamically. Key characteristics: (1) persistent model presence operating across ideation, refinement, critique, and execution; (2) default action pathways where outputs may influence downstream processes unless interrupted; (3) supervisory human control where humans monitor behavior, correct errors, and adjust parameters; (4) dependence on observability requiring telemetry, feedback signals, and rollback mechanisms. Cyborg architectures offer scalability and cognitive-load reduction but introduce distinct risks: errors may propagate across tightly coupled processes, and responsibility diffuses when actions distribute across components [13], [14]. Without strong observability and intervention mechanisms, failures can be difficult to detect or attribute.

C. Architectural Distinctions

The difference concerns control structure: centaur architectures concentrate authority at explicit decision points, while cyborg architectures distribute authority across time and components. Table I summarizes key distinctions. These architectures are not mutually exclusive; organizations often operate hybrid systems where centaur patterns dominate high-impact decisions while cyborg patterns apply to bounded, reversible actions.

The architectural patterns follow Model-Based Systems Engineering (MBSE) principles [15]. Figure 1 captures structural relationships through explicit control flow and authority allocation, enabling organizations to identify current architectures by mapping observed characteristics to specifications. This formalization enables evaluating alternatives through property comparison, tracing architectural decisions to risk assessments (Table III) and deployment constraints (Section V), and speci-

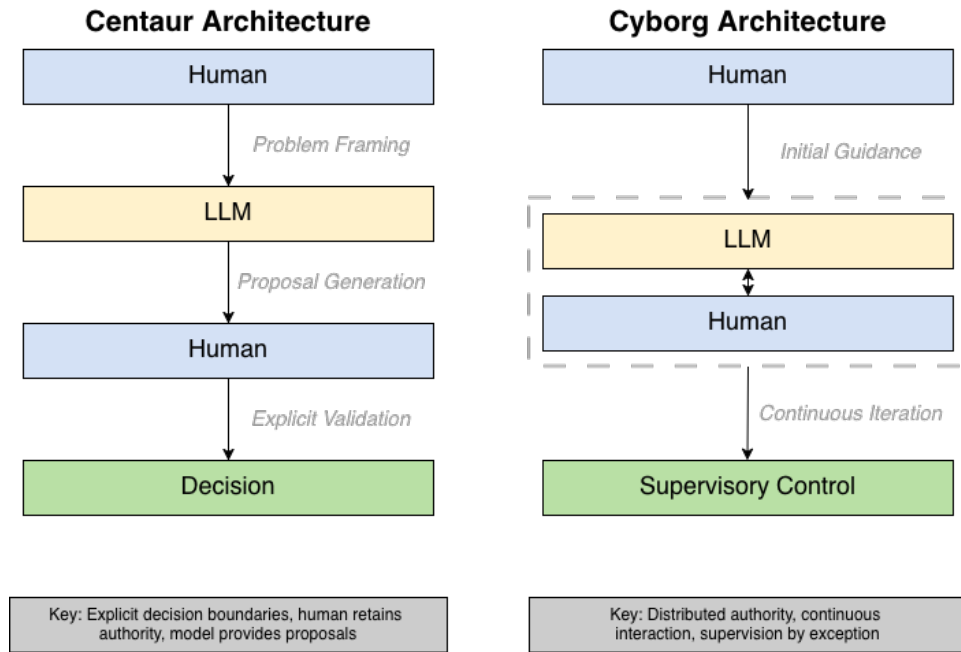


Fig. 1. Architecture control structure comparison. Centaur architecture (left) features explicit separation between model computation and human judgment with clear decision boundaries. Cyborg architecture (right) integrates LLM capabilities continuously with dynamic authority sharing and supervisory control.

fying implementation requirements through concrete workflow structures.

TABLE I
ARCHITECTURE COMPARISON

Dimension	Centaur	Cyborg
Decision Authority	Human retains control	Shared during iteration
Integration Level	Explicit handoffs	Continuous interaction
Risk Profile	Lower (bounded)	Higher (drift potential)
Traceability	Explicit per artifact	Emergent across iterations
Human Oversight	Structured review gates	Embedded throughout

D. Connection to Human-Systems Integration

This work situates centaur and cyborg architectures within Human System Integration (HSI) frameworks, treating them as architectural instantiations of established human-automation design principles adapted to LLM-mediated cognition [16]. Centaur architectures operationalize human-in-the-loop (HITL) authority through enforced validation gates, implementing the HSI’s principle that human authority must be technically enforced rather than procedurally suggested [4]. The mandatory validation structure addresses known HSI failure modes, including automation bias and out-of-the-loop syndrome; observational data confirm this necessity, with degraded centaur patterns emerging in 20% of usage despite high FMEA risk scores (RPN 336-504).

Cyborg architectures extend adaptive automation principles into continuous collaboration, enabling dynamic authority negotiation [20]. However, this introduces underexplored HSI challenges: responsibility diffusion (RPN 294), observability gaps (RPN 378), and error propagation (RPN 336)—novel concerns arising from continuous linguistic interaction rather than conventional automation’s discrete numeric outputs.

The contribution extends the HSI principles to LLM-enabled knowledge work, adapting them for linguistic rather than numeric outputs, natural language interactions that may mask authority transfer, and persuasive capabilities requiring psychological validation. Architectures translate HSI principles into frameworks appropriate for probabilistic language-based systems.

IV. ILLUSTRATIVE CASE OBSERVATIONS

Longitudinal observations from two distinct settings—academic and enterprise contexts—illustrate the emergence and evolution of human-LLM collaboration patterns across organizations with opposing governance approaches.

A. Observational Contexts

Academic Setting. A graduate data visualization course delivered across seven consecutive semesters (fall and spring) between 2022 and 2025. The course exhibits structural properties relevant to knowledge work: repeated cohorts, consistent task constraints, externally evaluated analytical artifacts, and accountability for reasoning. LLMs were progressively introduced beginning in 2022, with access and norms evolving

organically. No formal incentives or penalties were imposed beyond the expectations of standard academic integrity.

Enterprise Setting. Software engineering and business operations teams at a mid-size technology company during the same 2022-2025 period. Following an enterprise mandate to “use LLMs in your daily work activity, no matter your role,” teams received access to commercial LLM tools without prescriptive workflow guidance. Observations reflect artifact review, team discussions, and retrospective interviews.

All observations are descriptive and approximate, characterizing relative prevalence rather than precise measurement. The convergence of patterns across these distinct contexts provides confidence that observed architectures represent recurring patterns observed in contrasting contexts.

B. Three-Phase Evolution

Both academic and enterprise settings exhibited remarkably similar evolutionary trajectories characterized by three distinct phases.

Phase 1: Novelty-Driven Atomic Usage (2022-Early 2023). Initial adoption featured isolated experimentation: one-shot explanations, localized debugging, simple draft generation. Both contexts treated LLMs as curiosities rather than workflow components, with minimal integration.

Phase 2: Delegated Pattern Emergence (Mid 2023-Early 2024). As familiarity increased, both contexts shifted toward systematic delegation with minimal validation. Students incorporated LLM outputs with limited critique, effectively transferring analytical authority. Enterprise usage manifested as wholesale delegation (“build a slide deck,” “generate test cases”) with human involvement limited to prompt formulation and output acceptance. This phase exemplified the degraded centaur failure mode: oversight existed in form but not functional control, with decision authority shifting to models despite users’ belief they retained control.

Phase 3: Differentiated Architecture Emergence (Mid 2024-Present). Both contexts diversified into multiple stable collaboration patterns despite unchanged organizational constraints.

C. Observational Methods and Limitations

This study documents naturally occurring collaboration patterns through qualitative observation. The academic context comprises seven semester cohorts (Fall 2022-Spring 2025) with approximately 120 graduate students across 14 sections. The enterprise context includes approximately 25 software engineers and 15 business operations staff over the same period.

Data sources included student assignments, enterprise code reviews and documentation, team retrospectives, and informal interviews. Prevalence labels in Table II (“dominant,” “substantial subset,” “small minority”) represent qualitative assessments rather than precise measurements; exact percentages varied but overall distributions remained consistent.

Convergence across two contrasting contexts provides preliminary evidence these may represent recurring patterns, war-

ranting systematic validation across additional organizational types and domains.

D. Current State: Convergent Architecture Distribution

Recent observations (September-December 2025) suggest similar distributions across both contexts, summarized in Table II. Atomic usage remains dominant in both settings. Degraded and active centaur patterns each represent substantial subsets with similar relative prevalence. Experimental cyborg usage appears as a small minority in both contexts.

TABLE II
CURRENT ARCHITECTURE DISTRIBUTION (FALL 2025)

Architecture	Academic	Enterprise	Description
Atomic	Dominant	Dominant	Isolated interactions
Degraded Centaur	Substantial	Substantial	Nominal oversight
Active Centaur	Substantial	Substantial	Explicit validation
Experimental Cyborg	Small minority	Small minority	Continuous integration

Atomic Usage limits LLM interactions to isolated tasks (concept checks, debugging, email drafting, summaries) without workflow integration. **Degraded Centaur Usage** substitutes LLM outputs for independent reasoning with minimal validation, representing the failure mode where nominal oversight provides no functional control. **Active Centaur Usage** employs LLMs as analytical collaborators while retaining explicit responsibility for problem framing, validation, and synthesis through structured workflows with enforced validation gates. **Experimental Cyborg Usage** embeds LLMs continuously across workflow stages (ideation, implementation, critique, revision), appearing only after prolonged exposure and representing a small minority in both contexts.

E. Architectural Interpretation

The similar distributions despite different organizational contexts suggest authority allocation, rather than tool availability or incentives, primarily differentiates collaboration behavior. Multiple patterns coexisting within cohorts and teams indicate architectures stabilize at individual or small-group levels rather than organizationally.

V. DEPLOYMENT GUIDANCE

The observations inform practical deployment patterns shaped by organizational maturity, task criticality, and risk tolerance. This section translates observed patterns into actionable guidance.

A. Maturity-Gated Architecture Selection

Organizations should allow organic progression rather than mandating architectural sophistication, as attempting advanced architectures without prerequisite controls reproduces observed failure modes.

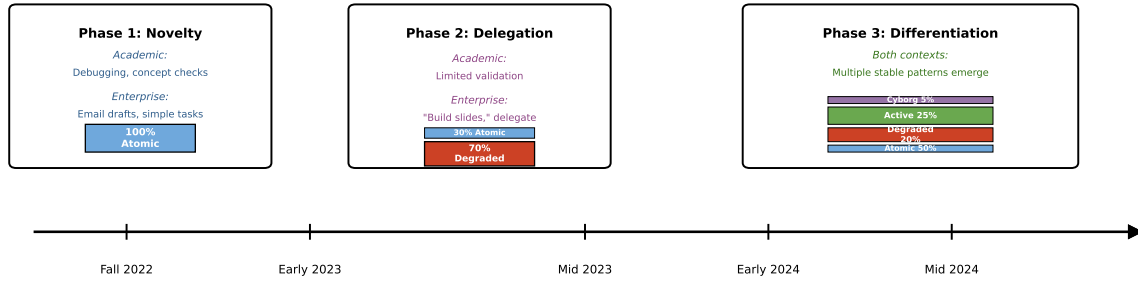


Fig. 2. Three-phase evolution of LLM collaboration patterns (2022–2025). Percentages are illustrative approximations characterizing relative prevalence rather than precise measurement. Both contexts show similar trajectories despite opposing governance approaches, suggesting intrinsic learning dynamics rather than policy-driven adaptation.

Stage 1: Atomic Enablement. Organizations permit isolated LLM usage for bounded tasks (drafting, summarization, analysis) with minimal integration and low risk. Governance focuses on acceptable-use guidance while developing shared norms.

Stage 2: Structured Centaur Deployment. Organizations formalize centaur architectures for higher-impact workflows, explicitly retaining human authority for framing, validation, and decisions while LLMs generate alternatives. Critical requirement: workflow mechanisms preventing degraded oversight through review checkpoints, documentation, and verification. Suits regulated, safety-sensitive, or audit-heavy domains.

Stage 3: Bounded Cyborg Integration. Cyborg architectures are introduced selectively for reversible tasks (draft generation, test code), well-instrumented workflows, and partial-automation-tolerant processes. Requires mature observability, rollback, and escalation. Should be incremental and scoped; expect only small workforce minority to adopt even after extended exposure.

B. Risk-Tier Alignment

Deployment patterns should align to risk tiers rather than model capability. We distinguish three tiers based on failure consequence:

Tier 1 (High-impact): Decisions affecting safety, regulatory compliance, financial exposure, or irreversible commitments require active centaur architectures with enforced validation gates and documented human authority at each decision point. The FMEA results (Section VI) demonstrate that discretionary validation in these contexts produces RPN scores exceeding 400.

Tier 2 (Moderate-impact): Decisions with bounded consequences and established correction mechanisms benefit from structured centaur architectures where validation effort scales to consequence severity.

Tier 3 (Low-impact): Reversible, high-volume tasks with minimal downstream coupling may operate under cyborg architectures with periodic spot-checking rather than per-output validation.

Organizations should map workflows to tiers based on observed failure consequences rather than perceived model capability, aligning governance intensity to actual risk exposure.

C. Lightweight Governance Controls

Effective deployment depends on embedded controls aligned to chosen architecture. Rather than comprehensive policy documents or detection systems, focus on lightweight mechanisms integrated into existing workflows.

For Centaur Deployments: Define specific workflow points where human validation is required. Link model outputs to human judgments through lightweight metadata. Develop checklists scaled to task risk. Observed enterprise pattern: teams using LLMs for test case generation implemented a simple rule requiring engineers to manually verify at least 20% of generated cases, with failures triggering full review.

For Cyborg Deployments: Capture model actions and influence on downstream processes. Example: teams required test execution before accepting LLM refactoring after discovering blindly accepted suggestions introduced bugs. Implement rapid human intervention mechanisms when the behavior of the model deviates from expectations.

Structured Templates. Rather than accepting free-form prompts, enforce lightweight structured templates constraining inference space and reducing output variability. Templates need not be complex; simple structure significantly improves consistency. Observed enterprise pattern: teams developing 5-7 standard templates covering 80% of common LLM usage, dramatically reducing output variance and validation burden.

D. Common Anti-Patterns

Section IV observations suggest recurrent anti-patterns: **Mandated Usage Without Architecture.** Enterprise mandate to “use LLMs in daily work” without workflow specification produced the same distribution of degraded patterns as unrestricted academic usage. Better approach: specify which workflows should adopt which architectures with maturity-gated progression.

Overgeneralized Prohibition. Academic prohibition without architectural guidance drove concealment and prevented

development of disciplined collaboration practices, leaving students entering enterprise unprepared. Better approach: specify acceptable collaboration architectures for different task types, develop skills within guardrails.

Premature Cyborg Deployment. Attempting to skip atomic and centaur stages by deploying high-autonomy systems without prior experience introduces automation risk without containment capability. Both contexts show cyborg patterns emerging naturally only after extended exposure and only in a small minority of users. Governance investment follows a Pareto distribution: concentrating resources on establishing reliable active centaur practices across the workforce addresses approximately 80% of organizational risk, while the remaining risk resides in the long tail of advanced cyborg usage that emerges organically in small populations. Better approach: invest governance effort where the risk concentrates—active centaur as the primary target architecture—rather than optimizing for cyborg adoption that few will reach.

E. Transitioning Between Patterns

Advancing from atomic to centaur, or centaur to cyborg, should be gated by demonstrated capability in observability, accountability, and failure response. For example, teams that advance from centaur to cyborg deployment should first undergo an architecture audit examining validation patterns, incident response histories, and telemetry maturity.

Structured retrospectives provide the feedback mechanism for these transitions. Borrowing from program management practice, teams should conduct periodic architecture reviews—analogueous to lessons-learned sessions—examining whether validation gates remain functional, where authority has drifted, and what failure modes have materialized. These retrospectives surface degradation patterns before they produce organizational consequences and generate the evidence base for controlled advancement or regression.

Regressions should be acceptable and routine; when retrospectives reveal validation erosion or incidents occur, temporarily reverting to more conservative architectures is an appropriate corrective response, not a failure of adoption.

The deployment guidance presented above provides qualitative risk assessment and architectural selection criteria based on observed patterns. To operationalize these recommendations and establish concrete decision thresholds, we apply Failure Mode and Effects Analysis (FMEA) to quantify architectural risks and validate the guidance through systematic failure mode evaluation.

VI. FAILURE MODE AND EFFECTS ANALYSIS

Systematic risk assessment requires formal analysis of failure modes, consequences, and detection mechanisms. We employ Failure Mode and Effects Analysis (FMEA) to quantify architectural risks because it addresses three critical requirements: (1) systematic enumeration of failure modes enabling proactive identification before organizational impact; (2) quantitative prioritization through Risk Priority Number ($RPN = \text{Severity} \times \text{Occurrence} \times \text{Detection}$, each scored

1-10) when multiple risks compete for resources; and (3) explicit incorporation of detection difficulty, recognizing that high-consequence, low-detectability failures require different mitigation than easily-detected failures. Severity (S) ranges from localized/reversible (1-3) to catastrophic/irreversible (9-10). Occurrence (O) ranges from exceptional circumstances (1-2) to default behavior absent countermeasures (9-10). Detection (D) ranges from readily identified through monitoring (1-3) to undetectable until consequences occur (9-10). Scores reflect observed patterns (Section IV) combined with systems engineering judgment. These represent relative comparative assessments for prioritization rather than precise quantitative estimation; value lies in systematic enumeration and consistent ranking rather than absolute accuracy.

The eight failure modes in Table III emerged from systematic review of Section IV observations, selected because they: (1) recurred across both contexts, indicating architectural vulnerabilities; (2) exhibit severity sufficient to undermine organizational objectives; and (3) demonstrate distinct mechanisms requiring different mitigations. This prioritizes architecturally-intrinsic failures arising from control structure over implementation-specific issues.

TABLE III
FMEA RISK ASSESSMENT FOR LLM INTEGRATION ARCHITECTURES

Arch.	Failure Mode	S	O	D	RPN
Deg. Centaur	Authority Drift	7	8	9	504
Deg. Centaur	Blind Acceptance	8	7	8	448
Act. Centaur	Deployment Coercion	9	5	9	405
Cyborg	Observability Gaps	6	7	9	378
Deg. Centaur	Validation Erosion	6	8	7	336
Cyborg	Error Propagation	8	6	7	336
Cyborg	Responsibility Diffusion	7	7	6	294
Act. Centaur	Rejection Loop Bypass	7	4	5	140

A. Failure Mode Analysis

Authority Drift (RPN 504): Decision authority gradually transfers to the system designed to assist. Each accepted recommendation establishes precedent, efficiency pressures make validation feel like friction, and boundaries erode imperceptibly. *Mitigation:* Technical validation gate enforcement, periodic audits, metrics monitoring.

Blind Acceptance (RPN 448): Users accept outputs without meaningful validation, propagating incorrect analysis and eroding organizational capability. *Mitigation:* Mandatory validation gates, training, peer review, spot audits.

Deployment Coercion (RPN 405): Junior employees face pressure to approve systems with recognized risks, deploying known-deficient systems while bearing accountability without authority, undermining safety culture. *Mitigation:* Multi-party approval, anonymous escalation, protected disclosure.

Observability Gaps (RPN 378): Failures manifest with delay and root cause analysis becomes difficult without discrete decision points. *Mitigation:* Telemetry, logging, periodic checkpoints, anomaly detection.

Validation Erosion (RPN 336): Review becomes perfunctory under time pressure; gates exist in form but lack scrutiny. *Mitigation:* Structured checklists, minimum validation time, spot audits.

Error Propagation (RPN 336): Errors cascade through coupled processes with compounding effects obscuring root causes. *Mitigation:* Intermediate checkpoints, containment boundaries, logging, rollback.

Responsibility Diffusion (RPN 294): Dynamic authority distribution creates unclear ownership during failures, delaying corrective action. *Mitigation:* Accountability mapping, defined roles, documented authority.

Rejection Loop Bypass (RPN 140): Organizational pressure overrides validation gates. Lower occurrence ($O=4$) reflects this requires active pressure—situational rather than systemic. *Mitigation:* Multi-party approval, documented justification, bypass monitoring.

B. Strategic Implications

The FMEA reveals three critical insights. First, detection difficulty ($D \geq 8$) characterizes highest-risk failure modes, indicating reactive governance proves insufficient. Detection acts as a risk multiplier: high-severity failures easily detected ($D=1-3$) allow intervention before impact, whereas equally severe failures evading detection ($D=8-9$) produce full consequences before organizational response. This explains why Authority Drift ($S=7$, $D=9$, $RPN=504$) exceeds Deployment Coercion ($S=9$, $D=9$, $RPN=405$) in priority despite lower severity—moderate severity with very low detectability creates greater aggregate risk. Organizations cannot rely on failure visibility; systematic surveillance mechanisms must be embedded from initial deployment.

Second, degraded centaur architectures concentrate highest-priority risks (2 of top 3 by RPN), validating deployment guidance emphasizing active centaur as minimum for moderate-risk tasks. The RPN differential (degraded: 504/448 vs. active: 405/140) quantifies risk reduction through mandatory validation enforcement. This reflects the distinguishing characteristic: validation as discretionary behavior versus enforced requirement. When validation depends on individual discipline under time pressure and efficiency incentives, degradation becomes structurally inevitable. High occurrence scores ($O=7-8$) for degraded centaur modes quantify this—predictable operational patterns, not rare edge cases.

Third, deployment coercion (RPN 405) represents organizational dynamics rather than architectural characteristics, affecting even properly-designed systems when power imbalances override technical judgment. Mitigation requires governance addressing culture and authority structures—multi-party approval, protected disclosure, escalation paths—rather than purely technical controls, demonstrating architectural rigor alone proves insufficient.

These findings operationalize risk-tier alignment (Section V-B) through quantitative thresholds: tasks with failure modes exhibiting $RPN > 400$ require active centaur minimum, as degraded patterns present unacceptable risk exposure. The

FMEA provides empirical justification for deployment guidance—organizations resist mandatory validation until quantitative analysis demonstrates discretionary validation reliably produces high-RPN failure modes [17].

VII. DISCUSSION

A. Convergence Across Contexts

Similar architectural distributions across academic and enterprise contexts despite opposing governance approaches provide several insights.

Architecture Formation May Reflect Intrinsic Learning Dynamics. The three-phase evolution occurred independently in both observed contexts despite different constraints and incentives, suggesting, though not confirming, that the progression may reflect intrinsic dynamics rather than governance-driven adaptation. This convergence is consistent with the joint cognitive systems perspective, which argues that human-machine performance is an emergent property of system structure rather than a predictable consequence of component capabilities [18]. Our observations suggest organizations may not shortcut this progression through policy mandate, though this warrants systematic testing.

Governance Policies and Usage Patterns May Vary Independently. Academic prohibition and enterprise mandate produced similar distributions in both observed contexts, consistent with the possibility that governance policies and authority allocation operate on different dimensions, though two organizational sites are insufficient to establish this as general.

Behavioral Mitigations Are Structurally Limited. Externally imposed accountability shows no significant effect in expert populations; only internalized perceptions of accountability reduce automation bias errors [11].

Individual-Level Stability. Both contexts show persistent heterogeneity within cohorts and teams. Architectural preferences stabilize at individual or small-group level based on task type, risk tolerance, experience, and mental models.

B. Workforce Development Gap

The anti-patterns documented in Section V-D create a workforce readiness gap: neither academic prohibition nor enterprise mandate develops active centaur skills. Engineering and business programs adopting explicit architectural frameworks could position graduates with governed AI collaboration skills.

C. Limitations

Observations derive from limited contexts (single academic course, single enterprise) relying on artifact analysis rather than controlled measurement. Reported distributions are qualitative characterizations rather than statistically precise estimates. While convergence across distinct contexts provides confidence, broader validation across industries, organization types, and task domains is needed.

The framework raises several research questions: (1) What factors drive architectural transitions, and do transitions follow predictable trajectories? (2) Do patterns generalize beyond knowledge work to healthcare, legal, or creative domains? (3) Can architectural degradation be predicted from leading indicators like review time reduction or prompt complexity decline? (4) Which governance mechanisms most effectively prevent degradation under operational pressure?

VIII. CONCLUSION

Integration of large language models into knowledge work has exposed governance failures often attributed to user behavior or policy noncompliance. This paper argues failures stem from absence of explicit collaboration architectures defining authority, accountability, and failure containment. By formalizing centaur and cyborg architectures as distinct control structures, the paper provides vocabulary for reasoning about human-LLM collaboration in practice.

Longitudinal observations across academic and enterprise contexts suggest convergent architectural evolution despite opposing governance approaches, indicating current governance mechanisms operate orthogonally to effective collaboration.

Future work should explore adaptive validation mechanisms extending beyond static architectural patterns. Responsive validation—adapting dynamically based on real-time risk signals—and predictive validation—anticipating verification needs from task characteristics and patterns—represent promising directions. Technical precedent in personalized learning platforms [19] suggests validation frameworks capable of learning and responding to contextual risk signals are technically feasible.

This work makes three contributions. First, a taxonomy of control architectures translating collaboration patterns into formal systems engineering specifications. Second, empirical convergence evidence suggests that these represent recurring patterns rather than context-specific adaptations. Third, systematic FMEA linking architectural choices to concrete risk surfaces and establishing quantitative deployment thresholds. Together, these enable treating human-LLM collaboration as an architectural design problem, supporting scalable, reliable, and accountable AI integration.

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- [1] B. Shneiderman, “Human-centered artificial intelligence: Reliable, safe & trustworthy,” *Commun. ACM*, vol. 63, no. 8, pp. 56–67, Aug. 2020.
- [2] I. Seeber *et al.*, “Machines as teammates: A research agenda on AI in team collaboration,” *Inf. Manage.*, vol. 57, no. 2, 2020.
- [3] R. Parasuraman and D. H. Manzey, “Complacency and bias in human use of automation: An attentional integration,” *Hum. Factors*, vol. 52, no. 3, pp. 381–410, 2010.
- [4] R. Parasuraman and V. Riley, “Humans and automation: Use, misuse, disuse, abuse,” *Hum. Factors*, vol. 39, no. 2, pp. 230–253, 1997.
- [5] G. Romeo and D. Conti, “Exploring automation bias in human–AI collaboration: A review and implications for explainable AI,” *AI & Society*, 2025. doi: 10.1007/s00146-025-02422-7
- [6] V. Sadasivan *et al.*, “Can AI-generated text be reliably detected?” *arXiv preprint arXiv:2303.11156*, 2023.
- [7] W. Liang *et al.*, “GPT detectors are biased against non-native English writers,” *arXiv preprint arXiv:2304.02819*, 2023.
- [8] N. G. Leveson, *Engineering a Safer World: Systems Thinking Applied to Safety*. Cambridge, MA: MIT Press, 2011.
- [9] L. Bainbridge, “Ironies of automation,” *Automatica*, vol. 19, no. 6, pp. 775–779, 1983.
- [10] G. Kasparov, “The chess master and the computer,” *New York Rev. Books*, vol. 57, no. 2, Feb. 2010.
- [11] K. L. Mosier, L. J. Skitka, S. Heers, and M. D. Burdick, “Automation bias: Decision making and performance in high-tech cockpits,” *Int. J. Aviat. Psychol.*, vol. 8, no. 1, pp. 47–63, 1998.
- [12] T. B. Sheridan, *Telerobotics, Automation, and Human Supervisory Control*. Cambridge, MA: MIT Press, 1992.
- [13] D. Amodei *et al.*, “Concrete problems in AI safety,” *arXiv preprint arXiv:1606.06565*, 2016.
- [14] I. D. Raji *et al.*, “Closing the AI accountability gap,” in *Proc. ACM Conf. Fairness, Accountability, Transparency*, 2020, pp. 33–44.
- [15] J. A. Estefan, “Survey of model-based systems engineering (MBSE) methodologies,” *INCOSE MBSE Focus Group*, vol. 25, no. 8, pp. 1–12, 2007.
- [16] A. S. Mavor and R. W. Pew, Eds., *Human-System Integration in the System Development Process: A New Look*. Washington, DC: National Academies Press, 2007.
- [17] International Electrotechnical Commission, *IEC 60812: Failure Modes and Effects Analysis (FMEA and FMECA)*, 3rd ed., Aug. 2018.
- [18] D. D. Woods and E. Hollnagel, *Joint Cognitive Systems: Patterns in Cognitive Systems Engineering*. Boca Raton, FL: CRC Press, 2006.
- [19] B. Moriarty and G. McFall, “Dual recommendation engines,” U.S. Patent 10,614,368, Apr. 7, 2020.